Research Areas for Analysis

Dimensions of Life Sciences Research

A structure for analysing life science research areas is provided by McCulloch AD, Huber G, Integrative biological modelling in silico: 4–19 in Bock et al. (2002), who describe the dimensions covered by systems biology research and its anticipated development, e.g. the integrative analysis of physiological function. Using the computational models of the heart as examples, they discuss three types of integration: structural integration, functional integration and synthesis. Systems biology covers several other orthogonal dimensions of integration and scale of analysis as well:

- Structure: gene, protein, macro complex, organelle, cell, network, tissue, organ, organism
- Function: regulatory, growth, metabolic, electrical, mechanical, transport, signalling
- Data to theory: empirical data, statistical modelling, predictive modelling, mathematical, ontologies, systems analysis, physical–chemical principles, theory
- Choice of model organism (and advantages for studies) or man: bacteria (rapid reproduction), monocellular eukaryotes (still relatively simple, rapid reproduction), mammalian cells (laboratory cell lines), fertilised eggs (developmental studies, but more difficult to obtain, maintain, artificial environment), model organisms (complex, more relevant), humans (highly complex and relevant, ethical and legal restrictions)
- Systematic to systems biology: time dependence, degree of connectivity of networks, interplay of experiment and computation, system perturbations, predictive capacity
- Reproductive and developmental processes of organisms
- Evolution and diversity at the population level
- Cross-organism interaction (disease interactions and transmission, food chain transmission, development of bacterial resistance, mutants of HIV, new influenza variants, etc.)

Choice of Area

Since bioinformatics and systems biology research has such a wide range of "dimensions", when deciding on research areas, how can researchers choose the most promising area or regions in this multidimensional space? Often the choice is made by building on previous work and capabilities at individual laboratories. The challenge for the future is to identify where bioinformatics and systems biology methods can be applied to usefully improve knowledge and applications, often expanding analysis to higher levels of integration of data. In many cases, the greatest insights are still obtained by "old-fashioned" direct observation of the living organism